

Extending the CAR trace to the fixed shell target

MathlibAnnex.CStarAlgebra.CAR.exists_state_extension_trace

theorem

The source trace first becomes a state on the existing target algebra.

Statement

Let A be the completed CAR algebra, the completion of the matrix system $M_{2^n}(\mathbb{C})$, and let τ be its normalized trace. Fix a representative shell family: for each chosen pure-GNS class j it specifies an automorphism α_j and partial isometries $w_{j,n}$ between the transported and root shells. Fix the links L_j chosen by this family in the selected atomic representation π . Throughout this Card,

$$T = C^*(\pi(A), \{L_j\}), \quad \iota : A \longrightarrow T, \quad \iota(a) = \pi(a).$$

Thus T is the already constructed concrete unital C*-algebra, and ι is its injective source map. The links and target are kept fixed. A state means a positive continuous complex linear functional taking 1 to 1. There exists a state ϕ on T satisfying $\phi(\iota(a)) = \tau(a)$ for every $a \in A$.

Assumptions

The representative shell family is fixed. No representation of the target is an additional input to this extension theorem.

Conclusion

The set of state extensions of τ to this same T is nonempty. Traciality on all of T and uniqueness of the extension are separate results.

The extension theorem supplies an existence witness. The later definition `traceExtension` chooses one such witness; it does not change T or create a universal completion.

Proof route

Transfer τ to the isometric copy $\iota(A)$, extend the bounded functional, and use normalization to obtain positivity.

▼ Proof steps

1. The injective unital C*-homomorphism ι is isometric. Consequently $\lambda(\iota(a)) = \tau(a)$ is well-defined and satisfies

$$|\lambda(\iota(a))| \leq \|a\| = \|\iota(a)\|, \quad \lambda(1) = 1.$$

The input bounds are `norm_trace_le` and `trace_one`, and injectivity is `shellFamilySourceHom_injective`.

2. Apply the state-extension theorem for an injective unital star homomorphism, `exists_state_extension_of_injective`, with source functional τ . Its Hahn–Banach step extends λ with norm at most 1; its normalized-contraction positivity argument makes the extension a state. Its restriction equality gives exactly $\phi \circ \iota = \tau$.

Main citations

- The stated existence or structural result — `MathlibAnnex.CStarAlgebra.CAR.exists_state_extension_trace`
- The fixed concrete target and source map — `MathlibAnnex.CStarAlgebra.CAR.shellFamilySourceHom`
- Faithfulness of the source embedding — `MathlibAnnex.CStarAlgebra.CAR.shellFamilySourceHom_injective`
- State extension along an injective unital embedding — `MathlibAnnex.Analysis.CStarAlgebra.exists_state_extension_of_injective`
- Normalization of the CAR trace — `MathlibAnnex.CStarAlgebra.CAR.trace_one`
- Contractive bound on the CAR trace — `MathlibAnnex.CStarAlgebra.CAR.norm_trace_le`

Supporting route explanation

Lean source signature (exact)

```
theorem exists_state_extension_trace (family : RepresentativeShellFamily) :
  ∃ ϕ : ShellFamilyTarget family →L[ℂ] ℂ,
    ϕ ∈ MathlibAnnex.Analysis.CStarAlgebra.stateSpace (ShellFamilyTarget family) ∧
    ∀ b, ϕ (shellFamilySourceHom family b) = trace b
```

In the source	Mathematical meaning
<code>family : RepresentativeShellFamily;</code> <code>ShellFamilyTarget family</code>	The fixed CAR shell family and once-chosen links L_j determine the same concrete target $T = C^*(\pi(A), \{L_j\})$ throughout.
<code>shellFamilySourceHom family;</code> <code>trace</code>	The faithful unital source map $\iota : A \rightarrow T$, $\iota(a) = \pi(a)$, and the normalized CAR trace $\tau : A \rightarrow \mathbb{C}$.
<code>∃ ϕ : ShellFamilyTarget family →L[ℂ] ℂ</code>	There is a continuous complex-linear functional $\phi : T \rightarrow \mathbb{C}$ satisfying both following requirements.
<code>ϕ ∈ ... stateSpace (ShellFamilyTarget family)</code>	This same ϕ is a state: positive and normalized by $\phi(1_T) = 1$.
<code>∀ b, ϕ (shellFamilySourceHom family b) = trace b</code>	For every source element $b \in A$, $\phi(\iota(b)) = \tau(b)$. Its values on the entire target are not assumed tracial here, and uniqueness is not this existence statement.

Further source notes: The linked proof invokes `exists_state_extension_of_injective`; that supplier is named in the proof, not in the displayed theorem signature.

Earlier published Card and PDF

▼ **Exact content identity**

Declaration: `MathlibAnnex.CStarAlgebra.CAR.exists_state_extension_trace`

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